

SWAROOP THIKANE

Electronics & Embedded Systems Engineer

+91 9766560792 swaroopthikane@gmail.com linkedin.com/in/swaroop-thikane-46a846296 Maharashtra

SUMMARY

Innovative and research-driven Electronics and Telecommunication Engineering student with hands-on experience in embedded systems, IoT, and power electronics. Skilled in C/C++, Embedded C, Verilog, VHDL, and prototyping using ESP32 and Arduino platforms. Completed industrial training at Kulkarni Power Tools Ltd., gaining exposure to IGBT-based welding systems. Targeting roles in Embedded Systems, Firmware Development, IoT Engineering, and Industrial Automation.

EXPERIENCE

Trainee

Kulkarni Power Tools Ltd. 12/2024 Shirol

- Contributed to the **development and inspection** of PCB layouts for **welding control systems** using advanced **industrial design software**.
- Studied and evaluated** the behavior of **IGBTs and MOSFETs** in welding circuits to understand **switching dynamics and energy efficiency**.
- Handled** IGBT-based welding machines under guidance to **assess load responses** and verify system reliability.
- Assisted in prototype trials**, ensuring **stable voltage output** and consistent circuit performance during testing.
- Acquired hands-on experience** in **PCB manufacturing and assembly operations**, identifying ways to improve **component alignment and process accuracy**.

LEADERSHIP & VOLUNTEERING

IEEE SB Treasurer

07/2025 - Present
Managed funds, sponsorships, and coordinated technical events for **200+** members.

COMSA President

06/2024 - 06/2025
Led **40-member** student body, organizing technical workshops and hackathons to foster innovation.

PROJECTS

KRISHICOBOT- Agriculture Assistive Robot

- 07/2025 - Present
- Built an agri - automation robot integrating a UR5 robotic arm and vision sensors for fertilizer refilling and fruit sorting.
 - Enhanced operational efficiency by 30% through autonomous crop detection, fertilizer dispensing, and defective fruit identification.

IoT-Based Pearl Farm Health Regulator

- 12/2024 - 09/2025
- Engineered an IoT-enabled system using ESP32 and cloud sensors to monitor pH, TDS, and oxygen levels in pearl farms.
 - Automated algae feeding through microcontroller-driven actuators, securing a ₹6 Lakh IIT Bombay fellowship, design patent, and IEEE publication acceptance.

Filament Maker

- 02/2024 - 07/2024
- Developed a recycling system converting PET bottles into 3D printing filaments using stepper motors, drivers, and relays.
 - Contributed to sustainable prototyping, reducing plastic waste by 40% and enabling low-cost filament production in the R&D lab.

Smart Bag System

- 08/2023 - 01/2024
- Designed a multifunctional RFID and GPS-based smart bag offering real-time location tracking and inventory management.
 - Developed the prototype using GSM, GPS, and RFID modules, resulting in a published utility patent for innovative functionality.

EDUCATION

B.Tech - Electronics & Telecommunication Engineering

D.K.T.E's Textile & Engineering Institute

11/2022 - 06/2026 Ichalkaranji

CGPA

7.82 / 10

LANGUAGES

English Proficient Hindi Proficient Marathi Native

KEY ACHIEVEMENTS

Fellowship Award

Received ₹6,00,000 grant under **TIH CHANAKYA Fellowship, IIT Bombay (2024)** for IoT-based Pearl Farm Health Regulator project.

Design Patent Granted

IoT-Based Pearl Farm Health Regulator (Design No. 454051-001)

Utility Patent Published

Smart Bag System for Enhanced Safety, Organization, and Educational Engagement (App. No. 202421041453)

Winner - Mini Project Competition, DKTE (2024)

Winner - Ideathon, DKTE | Best Idea Presenter | Finalist at NRIT Ideathon

RESEARCH & PUBLICATIONS

IoT-Based Pearl Farm Health Regulator - Accepted for publication in ICFT 2025 (IEEE)

IoT-Based Pearl Farm Health Regulator - Accepted for publication in ETET 2025 (Scopus)

CERTIFICATIONS

Introduction to Internet of Things - IIT Bombay (Coursera), Feb 2025

Principles of Management - NPTEL, Apr 2025

CLA: Programming Essentials in C - Cisco, Mar 2025

SKILLS

Programming & Software Development

C, C++, Embedded C, Java, Python, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Visual Studio Code, Eclipse IDE, Debugging & Testing

Embedded Systems & Firmware Development

Microcontroller Programming (Arduino, ESP32, PIC, ARM7-LPC2148), RTOS, Interrupts, UART/SPI/I2C, ADC/DAC, Embedded C Standards, Peripheral Integration, Memory Management

IoT & Communication Technologies

Internet of Things (IoT), MQTT, GSM/GPRS, GPS Module Integration, RFID Systems, Sensor Interfacing, Node MCU, Cloud Connectivity (Thing Speak, BLYNK, AWS IoT), Wi-Fi/Bluetooth Communication, Edge Computing, Real-Time Data Monitoring

Electronics Design & Hardware Development

PCB Design (KiCad, EasyEDA), Circuit Simulation (Proteus), Analog & Digital Electronics, Soldering & Prototyping, Power Electronics, Motor Control Systems, Stepper/Servo Motor Drivers, Relay Control, Signal Processing, Hardware Debugging, Oscilloscope & Multimeter Usage

Automation, Robotics & Control Systems

PLC Programming, SCADA, Mechatronics, UR5 Arm Integration, Sensor Calibration, Actuator Control, Control Systems, Industrial Automation, Process Monitoring, Robotic Vision, Feedback & PID Control

Research, Project & Management Skills

Technical Documentation, Design Thinking, Project Management, Rapid Prototyping, Cross-Functional Collaboration, Technical Writing, Research & Publication, Innovation & Patenting, Leadership & Team Coordination, Event Organization, Presentation & Communication